



Solution Guide: Compound Commands

Completing this activity required the following steps:

- Creating a directory and automatically copying log files to it with one command.
- Finding a list of executable files in the home folder and saving it to a text file inside your directory with one command.
- Saving an edited list of the 10 most active processes to your directory with one command.
- Creating a list of home folders and users with a UID less than 1,000 and saving it to your directory, all with one command.

Log into the lab environment with the username `sysadmin` and password `cybersecurity`.

Create a research directory and copy all system logs as well as the `shadow`, `passwd`, and `hosts` files in one long command.

- Run `mkdir -p ~/research && cp -r /var/log/* /etc/passwd /etc/shadow /etc/hosts ~/research`. **-NOTE:** The `-p` option will create the directory if it doesn't exist, but won't give an error if it does.

We'll use `&&` to combine the two following commands together:

- `mkdir` to make our directory.
- `cp` to copy our files to our new directory.
- We also need to use `sudo` because we are making copies of sensitive `/etc` files.

Type the solution into the command line:

- `mkdir -p ~/research && sudo cp -r /var/log/* /etc/passwd /etc/shadow /etc/hosts ~/research`

Syntax breakdown:

- **`mkdir -p ~/research`:** Creates our directory if it does not already exist.
- **`&&`:** Completes the second command if the first command is successful.
- **`sudo cp -r /var/log/* /etc/passwd /etc/shadow /etc/hosts`:** Chains together a number of files—as many as we want—to copy (recursively).
- **`~/research`:** Output directory that we created with the first command.

Remember the command we discussed at the beginning of class, namely `file $(find / -iname '*.txt' 2>/dev/null) > ~/Desktop/text_files ; tail ~/Desktop/text_files`.

- This command is an example in which `&&` might be better to use than `;` before we issue the `tail` command. This way, the file is completely written before we open it.

Create a list of all files with `executable` permissions and save it to a text file in the `research` folder using one long command.

- Run `sudo find / -type f \( -perm -u=x -o -perm -g=x -o -perm -o=x \) > ~/research/exec_lst.txt`.

This task only requires using one command, along with an output redirect to direct the list into a file that we specify. Again, we need to use `sudo` to search the entire system.

- Run `find / -type f \( -perm -u=x -o -perm -g=x -o -perm -o=x \) > ~/research/exec_lst.txt`.

Syntax breakdown:

- `sudo find` searches the entire directory.
- `/` starts our search in the `root` directory.
- `-type f` searches for objects that are files (not directories).
- `( -perm -u=x -o -perm -g=x -o -perm -o=x \)` searches for any files that have the executable bit set for owner, user, or other.
- `> ~/research/exec_lst.txt` redirects the list returned by `find` to a text file.

Bonus

Create a list of home folders along with user info from the `passwd` file. Only add the user info to your list if the UID is greater than 1,000.

- Run `ls /home > ~/research/users.txt && cat /etc/passwd | awk -F ":" '{if ($3 >= 1000) print $0}' >> ~/research/users.txt`.
- We will again need to use `awk` to parse the output of the `passwd` file.
- Type the first part of the command `ls /home > ~/research/users.txt &&`.
 - This command creates a list of the home folders and saves it. Then, we are using the `&&` to make sure that this command completes before we add more to that file.

Type out the next part of the command `cat /etc/passwd | awk -F ":" '{if ($3 >= 1000) print $0}'`.

- `cat /etc/passwd` gives us the entire contents of the `passwd` file.
- `|` sends those contents to our next command.
- `awk` allows us to parse the output.

- `-F ":"` changes the delimiter that `awk` is using to parse input. By default, `awk` uses white space to divide lines of text, but here we are changing it to a colon because items are separated by a colon in the `passwd` file.
- `'{if ($3 >= 1000) print $0}'`: This is an `if` statement inside of `awk`. It says, if the third item given is greater than 1,000, print `$0`, which is the entire line.
 - Remember that `awk` assigns each item a number and the number 0 is assigned to the entire line.

Run the entire command `ls /home > ~/research/users.txt && cat /etc/passwd | awk -F ":" '{if ($3 >= 1000) print $0}' >> ~/research/users.txt.`
